

- 5 (a) Show that the curve with Cartesian equation

$$x^2 - y^2 = a,$$

where a is a positive constant, has polar equation $r^2 = a \sec 2\theta$. [3]

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The curve C has polar equation $r^2 = a \sec 2\theta$, where a is a positive constant, for $0 \leq \theta < \frac{1}{4}\pi$.

- (b) Sketch C and state the minimum distance of C from the pole. [3]

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- (c) Find, in terms of a , the exact value of the area of the region enclosed by C , the initial line, and the half-line $\theta = \frac{1}{12}\pi$. [You may use any result from the list of formulae (MF19) without proof.] [4]

[illegible]